Multi-Camera Federated Edge-AI Based Anti-Spoofing Smart Monitoring System

A thesis submitted in partial fulfilment of the requirements for the degree of

Master of Technology

by

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Dedicated to my parents

I lovingly dedicate this thesis to my beloved parents, whose unconditional love, endless support, and constant encouragement have been the foundation of my academic journey.

Abstract

The surveillance and authentication are widely dependent on face recognition system but such systems are vulnerable to spoofing technique like replay videos and printed images. This paper represents a multi-camera edge-based anti-spoofing system that uses real-time identification using a lightweight deep learning model. This model integrated video-based temporal information with image-based facial attributes to increase model robustness. The proposed technology executes data on the local system that decreases latency and protects privacy for multiple face detections. The proposed model represents that hybrid data integration increase the model performance and detection accuracy. The suggested model indicates a flexible and effective result for real-time smart monitoring applications.

Keywords: Deep learning, Edge AI, Face Anti-Spoofing, Federated Learning, Liveness Detection, Real-Time Monitoring

Declaration

I certify that

- The work contained in the dissertation is original and has been done by myself under the general supervision of my supervisor.
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All helps received by him from various sources have been duly acknowledged. No part of this thesis has been submitted elsewhere for award of any other degree.

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Chapter 1

Introduction

Face recognition technology has become increasingly prevalent in banking, smart surveillance, mobile authentication, and security systems. It doesn't need any physical contact and it is fast and easy to use. Traditional monitoring technology commonly based on physical credentials such as keys or access cards, these are subject to loss or misuse and creating vulnerabilities. Subsequently, there has been a shift toward biometric-based authentication methods [14]. Widely used biometric modalities include facial recognition [18],[21]along with fingerprint, iris, palm print, ear, voice [11], finger vein, and gait recognition [19].

In case of fingerprint recognition system it is easy to implement and its performance can decrease due to variations in skin condition. Although Iris recognition offers high accuracy but it requires dedicated hardware, making it less practical for large-scale deployment. Similarly, finger vein recognition offers strong security, but it requires special devices and direct physical contact, which can reduce efficiency and make real-time system performance slower. With the rapid development of imaging devices and surveillance infrastructure, facial data acquisition has become highly convenient. As a result, face recognition (FR)[16],[2],[25]has emerged as a preferred solution due to its contactless nature, ease of integration, and reliable performance. It is widely adopted in applications such as access control in transportation hubs, telecom services, financial authentication, and workforce management systems [17].

1.1 Challenges and Issues

However, despite the rapid progress of face recognition (FR) technologies, significant security challenges have emerged. The primary issue with biometric security is the identification of fake face that tricks the system so that it cannot recognize the originality of a face, such technique commonly known as face spoofing. Early in 2024, number of incidents were reported in different cities including Shanghai and Xuzhou, that reveals criminals using silicone face masks to impersonate individuals and involved in theft activity. These kind of events clearly show that spoofing techniques are becoming more practical and harder to detect. This report according to China Central Television (CCTV) Finance.

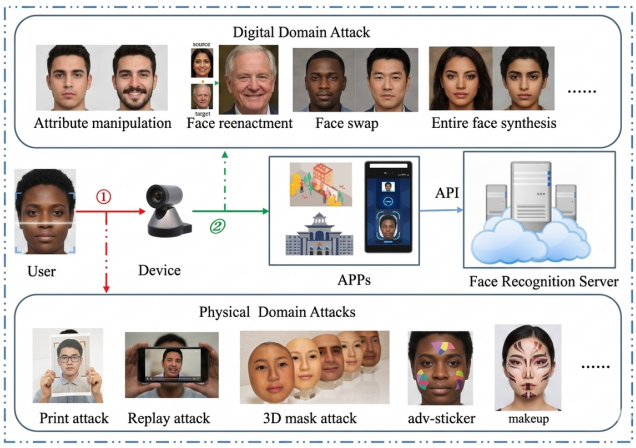


Figure 1-1: Face Recognition System Vulnerabilities for physical and digital attacks

The fast progress of generative adversarial networks (GANs) [3] has significantly enhanced the capabilities of face deepfake technologies. As a result, manipulated images and videos have become highly realistic [24], making them much more challenging to detect with the naked eye or even automated systems. As shown in the upper part of Figure 1, face deepfakes are based on deep learning (DL) and enable deliberate manipulation of facial content in images and videos, including face attribute modification [6], face re-enactment, face swapping, and complete face synthesis [29].

In practical deployments, however, face recognition systems (FRSs) typically acquire facial data directly through sensors and process it without additional validation, making them more susceptible to such sophisticated manipulations. While tampering with digital data during transmission is relatively complex, attacks in the physical domain are easier to execute. In real-world environments, FRSs are especially vulnerable to presentation attacks (PAs), such as print attacks,

replay attacks, 3D mask attacks, physical adversarial examples, and disguises using makeup or tattoos [29]. Therefore, the focus is primarily placed on understanding presentation attacks and exploring effective detection methods within the physical domain.

1.2 Motivation of the Research

With the rapid development of face recognition systems such as smart surveillance, attendance management, and secure authentication, ensuring the authenticity of a person has become a critical challenge. Traditional systems mainly focus on identifying faces but often fail to verify whether the input is from a real person or a spoof attack such as printed photos, replayed videos, or digital manipulations. The recent improvements in deepfake technologies have further increased the risk of identity fraud, making existing systems more vulnerable to identity fraud.

In addition, many existing systems still depend on manual identification. For example, in CCTV surveillance, identifying whether a person is known or unknown often requires human effort, where individuals must manually search and identify faces. Therefore, such processes take time, less efficient, and prone to human error.

The maximum anti-spoofing system require high computational models and centralized processing systems, that is a certain drawback to implementation in real-time applications and edge devices. More over, a lot of systems have struggled to detect multiple faces at the same time from various camera angles and they find it difficult to detect in changing environmental circumstances. These challenges clearly indicate the necessity for an anti-spoofing solution.

To overcome these limitations, this work presents a Multi-Camera Federated Edge-AI based smart monitoring system for face anti-spoofing as well as smart monitoring. The proposed framework supports a hybrid dataset that combines both image-based and video-based inputs, allowing the model to learn more diverse features and improve its ability to handle different types of spoofing attacks. A lightweight deep learning model was built on MobileNet to ensure fast inference and low computational overhead, which makes it well-suited for implementing on edge devices with limited resources.

The system can detect multiple faces for real-time detection, enabling reliable performance even in crowded scenarios. Furthermore, face detection is tightly

integrated with liveness assessment, which helps in improving classification accuracy while reducing false positives and false negatives. This architecture reduces dependence on centralized servers by using an edge-AI model, which lower the latency and permits faster the response time.

The effectiveness of the proposed approach is validated through extensive experiments on image-based, video-based, and hybrid datasets. The results demonstrate that the system achieves strong performance and maintains high reliability when applied to real-world environments. Furthermore, the proposed architecture is scalable and can be extended to multi-camera environments and integrated with federated learning techniques for privacy-preserving model updates.

1.3 Research Objectives

The main objective of this project to create a reliable and effective anti-spoofing solution for smart surveillance application. The following goals are implemented in order to fulfil this main objective:

To create a multi-camera monitoring system that can concurrently identify and validate several faces from different camera angles and in a range of environmental.

To overcome the vulnerabilities of current face recognition systems by creating a system that can identify both digital (such as deepfake) and physical (such as print, replay, and 3D mask) presentation attacks.

To develop a lightweight deep learning model that integrate image-based and video-based temporal information into a hybrid dataset that is suitable for implement on resource-constrained edge devices, ensuring low latency and real-time performance.

To make sure data privacy by implementing federated learning by processing and analyzing facial data locally at the edge, therefore it is minimizing the risks of centralizing sensitive biometric informations.

1.4 Thesis Contributions

The following are the main contributions for this study of smart surveillance and face anti-spoofing:

A Novel Multi-Camera Edge-AI Framework: The main aim is to design and implementation of a novel smart monitoring system that generate an effective, scalable, and privacy-preserving anti-spoofing solution by combining edge computing, federated learning principles, and a multi-camera arrangement.

A Lightweight Hybrid Deep Learning Model: This model based on the MobileNet architecture, the thesis suggests a lightweight deep learning model that successfully use a hybrid dataset that combines both temporal information from videos with image-based facial features. The computational processing of current centralized systems are overcome by this architecture, which is designed for low-latency inference on edge devices.

Real-Time Multi-Face Detection: The system is design to reliably recognize and detect multiple faces in a single frame makes it useful for large-scale surveillance deployments in busy public areas.

Comprehensive Problem Analysis: The thesis represents a full analysis of present problems in face anti-spoofing, including deepfake generation, physical presentation attacks, and the limitations of existing detection methods, thereby explaining the need for the proposed solution.

1.5 Thesis Organization

This thesis is organized into six chapters, structured as follows:

Chapter 1: Introduction

This chapter describes face recognition technology and the rising risk of spoofing attacks.

Chapter 2: Background Details

This chapter deals with basic ideas of proposed work such as deep learning, spoofing attack types, federated learning, edge computing, and face recognition systems.

Chapter 3: Literature Survey

This chapter evaluates existing face anti-spoofing techniques and evaluates their strengths and shortcomings.

Chapter 4: Proposed Approach

This chapter presents the detailed design of the Multi-Camera Federated Edge-AI Based Anti-Spoofing System. It describes the lightweight MobileNet-based architecture, hybrid data integration strategy, and real-time multi-face detection mechanism.

Chapter 5: Experimental Results

This chapter reports the simulation setup, performance evaluation, and comparative analysis.

Chapter 2

Related Work

2.1 Introduction

Recent advances in face anti-spoofing focus on improving robustness, generalization, and efficiency using deep learning models. Early research concentrated on identifying basic spoofing attempts Using manually created characteristics and conventional image processing methods[5]. Although these techniques worked good in a controlled environment, they were not well performed in real-world situations.

This [7] study a surveillance-oriented face anti-spoofing framework designed for real-world environments with multiple cameras. The aim of the research is to improve detection performance in difficult situations, including low-resolution inputs, pose fluctuations, and fluctuating lighting. The authors successfully identify between real and fake faces by integrating temporal information with deep learning-based feature extraction. This finding, which makes use of robust feature learning and multi-frame analysis, performs well in surveillance scenarios.However, the method's dependence on centralized processing and large computational resources may limit its use in edge-based systems.

This study [1] looks at thousands of papers from several databases to provide a comprehensive analysis of deepfake and anti-spoofing research. The authors find the growing significance of deep learning techniques in this domain by analyzing major research trends, approaches, and difficulties in deepfake detection. This paper also represents major themes such as feature extraction, face recognition, and security concerns, offering useful insights for further research directions. Still,

the study does not propose a workable anti-spoofing technique for real-time applications because it is mostly bibliometric and analytical in nature. Therefore, it is free of edge deployment considerations, real-time system design, and experimental validation—all of which are essential for realistic monitoring systems.

In this paper [22] provides a SIFT-based technique that utilizes geometric transformation estimation and duplicate region recognition to detect copy-move picture forgeries. This method is used well to identify tampered sections and is strong to scaling and rotation. But it has limited keypoint detection, it is less productive in low-texture areas and computationally costly. Furthermore, real-time evaluation and video-based monitoring are not supported by the approach, which is only for offline image forensics.

The International Journal of Advanced Computer Science and Applications, Alrethea and Ahmad [4] proposed a hybrid approach combining handcrafted chainlet features with convolutional neural networks. Where as their approach improves feature representation, it still relies on manual preprocessing, which restricts flexibility in the face of unknown threats.

Another work by Ikram [15] presented an attention-guided deep neural network and a Fourier transform. Here collecting information from both the spatial and frequency domains, the model greatly increases detection accuracy in a variety of scenarios such as edge deployment but still the architecture is comparatively complicated and computation is costly.

A spoof detection method based on DCGAN was introduced in [23], where the training dataset is improved and classification accuracy is increased through the use of generative adversarial networks. Although this approach increases performance but it has disadvantage of requiring a huge amount of training data and more processing time.

In a study Xing et al. [27] provided a detailed analysis of deep learning techniques for face anti-spoofing. According to their investigation, the great feature extraction capabilities of convolutional neural network (CNN)-based techniques make them popular. However, these models frequently have problems generalizing when subjected to different environmental conditions or attack types that have never been observed before.

Radad et al. [22] suggested a CNN framework that is encoder-driven and makes use of grayscale structural texture information. This method improves detection result by using both deep representations and manually created textural

features. However, this combination results is increased processing needs and model complexity.

Recently, a dynamically aware heterogeneous network designed to improve multimodal feature representation was presented [28]. While this method successfully combines data from various modalities, it requires a complex design and significant processing power.

In addition, Zhou et al. [32] looked into adversarial attacks on anti-spoofing and facial recognition systems. Their findings point out the need for more improved and modular detection techniques by showing that existing models are still vulnerable to effectively built attacks.

Another emerging direction represents multimodal and interpretable models, where large language models are integrated with visual features for better generalization [30]. These approaches improve clarity but are not suitable for real-time edge deployment due to high computational needs.

In [12], hybrid physical and digital attack detection techniques were explored. It shows that improves generalization across attack types the simulated spoofing clues are used. Despite their effectiveness, these methods mostly rely on the creation of synthetic data.

2.2 Summary

Overall, current research shows significant improvements in detection accuracy, but there are still significant drawbacks, such as high processing costs, a lack of real-time capacity, and poor generalization to unknown attacks. Therefore, there is a need of lightweight, hybrid, and edge-based system. The proposed system addresses these challenges by combining video and image data with an efficient deep learning model suitable for real-time multi-camera environments.

Chapter 3

Proposed Approach

3.1 Introduction

The system is designed as a multi-camera, edge-based anti-spoofing monitoring framework that can detect real and fake faces in real time. The overall working process is shown in Fig. 3-1 and is explained step by step below.

First, multiple cameras continuously capture video data from different locations. These inputs are combined as aggregated video feeds, which are then sent to the edge AI processing unit. Processing at the edge reduces delay and improves data privacy since the data does not need to be sent to a central server.

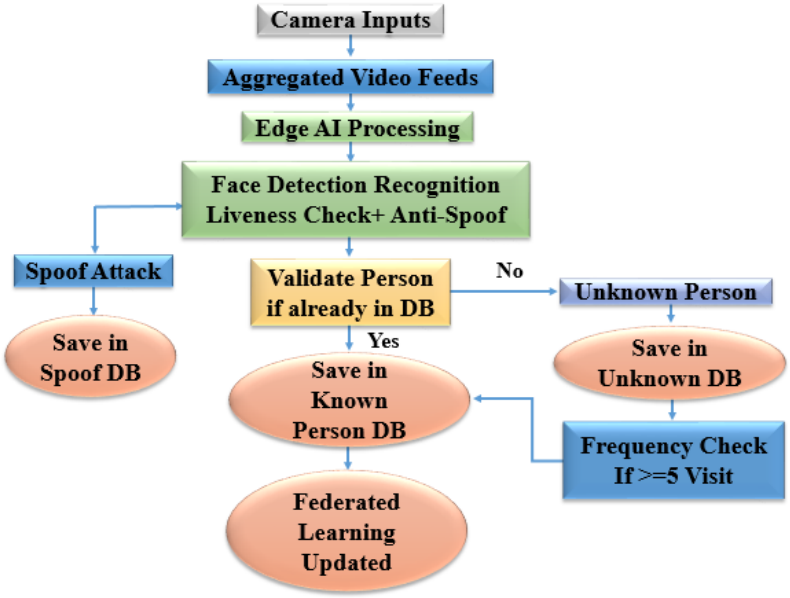


Figure 3-1: Proposed framework for Smart Monitoring System

In the next stage, the system performs face detection and recognition, followed by liveness and anti-spoofing checks. This step determines if the identified face belongs to a real person or represents a spoof attack, which can be a printed image or video replay.

When a spoof attack is identified, data is immediately stored in the spoof database for future reference and analysis. If the face is identified as real, the system checks whether the person already exists in the database.

If the person is already registered, their data is updated in the Known Person Database. If the person is new and not registered in the database, the information is stored in the Unknown Person Database. Moreover, the system is capable of using a frequency-based monitoring technique that tracks repeated appearances over time, for example, when the number of visits is ≥ 5. This makes it possible to identify the frequency of visits. If the frequency of visits of a person is more than five times, the information of that person can be stored in the Known Person Database, thereby supporting the detection of potentially suspicious or unusual activity patterns.

However, the system implements federated learning, which means that local data from multiple edge devices is used to update the model without exchanging raw data. This preserves privacy while improving the model's performance.

3.2 Dataset Description

The proposed anti-spoofing system is trained and evaluated using two different kinds of datasets: an image-based dataset and a video-based dataset. Both datasets are taken from publicly available sources on Kaggle and are organized to support hybrid learning for improved performance.

The primary component of the training data is the video dataset. It is arranged into separate directories for training and testing, where each sample represents a short video clip. The training folder contains 1168 video files along with a corresponding label file (`label.csv`), which includes two fields: the video file name (`fname`) and the liveness label (`liveness_score`). A value of 1 represents a real (live) face, whereas a value of 0 represents a spoof attack. Two additional folders, namely public and private test sets, are used for evaluation. During preprocessing, frames are extracted from each video, resized to a fixed resolution, normalized, and used as input to the model for capturing temporal variations

across the frames.

The second dataset is an image-based dataset included to provide greater robustness to the model. This dataset is divided into two main folders, namely `Live` and `Spoof`, where each folder contains cropped facial images representing real and fake samples. In this data set the image data helps the model to learn facial details, such as small texture changes and subtle patterns that can be useful for detecting spoofing attempts.

On the other hand, the Video sequences provides information about how facial features are changing with respect to time. The model can make more useful predictions under different real-world conditions with the integration of both spatial and temporal information. In real-world conditions, changes in lighting, differences in camera quality, and situations involving multiple faces. The well-organized dataset structure helps the data be easier to manage and load during training. Overall, this approach makes the system more scalable and suitable for real-time use in practical environments. [7, 22, 27].

3.3 Data Preprocessing

During data preprocessing, the input data has collected from two sources: videos and images. Since the videos and images have different formats and structures, they need to be converted into a common format. This process is used to maintain data consistency and ensures the input data is properly prepared for training and testing.

First, each video in the dataset is divided into a fixed number of frames. After that a few representative frames are selected from different parts of the video rather than using every frame, which can increase the computational load. In this way the model used to capture changes in facial appearance over time while keeping the preprocessing phase faster and more efficient.

Second all extracted frames and images are then resized to a specific resolution of 160×160 pixels. This makes all inputs have the same dimensions for training which speeds up processing and reduces memory usage. After resizing, the pixel values of the images are normalized from their original range of 0–255 to a range between 0 and 1. This helps the model to learn more effectively by reducing large variations in input values. The processed images are then converted into tensor format. This standard input format is required by deep learning frameworks

such as PyTorch, where the image data is rearranged so that the color channels come first, followed by the height and width dimensions.

For the video dataset, the selected frames are combined by calculating their average, creating a single representation of the video. This helps to keep the overall information from the video while reducing the amount of data the model needs to process. Each sample is then assigned a label according to its category: 1 for a real or live face and 0 for a spoof or fake face. These labels allow the model to learn the difference between genuine and spoofed faces.

Finally, the image and video data are combined to create a hybrid dataset. The image data helps the model learn detailed facial features, such as texture and subtle patterns, while the video data provides information about facial movement and changes over time. By using both types of data, the model can better understand whether a face is real or a spoof. By combining both types of data, the model can achieve improved performance under different real-world conditions.

Overall, these preprocessing steps ensure that the data is uniform, consistent, and optimized for training, which improves the performance and reliability of the proposed face anti-spoofing system.

3.4 Model Building

The suggested system is implemented as a multi-layer intelligent monitoring framework that integrated face detection, anti-spoofing, and face recognition. The complete processing pipeline is described in Fig. 3-2. The system receives video input from multiple cameras and processes those captured frames through different AI-based modules.

In this proposed system, the face detection and face recognition phase use pre-trained models, whereas the anti-spoofing phase uses the MobileNetV2-based model, which was developed and trained as part of this research.

3.4.1 Face Detection Using Pre-trained YuNet

In this suggested system the first phase is face detection. The system uses the YuNet face detection model, which is a lightweight CNN-based face detector designed for fast face detection, which is provided through the OpenCV model zoo.

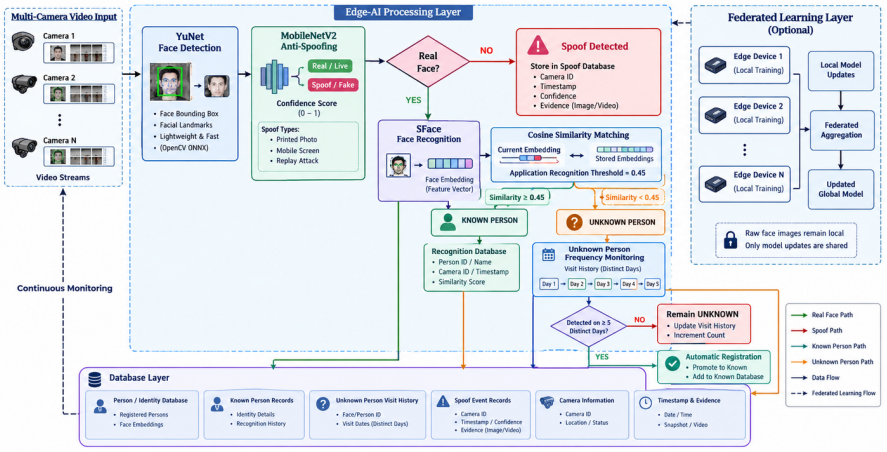


Figure 3-2: Proposed Multi-Camera Edge-AI Face Authentication, Anti-Spoofing, and Frequency-Based Identity Management Pipeline for Smart Monitoring System

It can be used directly with OpenCV for image- and camera-based face detection.

In the proposed method, YuNet uses bounding boxes to determine the facial area after receiving the camera frame. Then the detected facial region passed through the anti-spoofing stage.‘

The use of a pre-trained YuNet model eliminates the need to train a separate face-detection model using the project’s own dataset. Therefore, the work performed in this stage consists primarily of model integration, parameter configuration, camera-frame processing, and testing rather than model training.

The face detection stage can be represented as:

$$I_t \xrightarrow{\text{YuNet}} F_t \tag{3.1}$$

where:

- I_t represents the input camera frame at time t.
- F_t represents the detected face region at time t [26].

3.4.2 Anti-Spoofing Using MobileNetV2

The proposed system uses a lightweight Convolutional Neural Network (CNN) inspired by the MobileNet architecture to perform face anti-spoofing effectively in

real time. The main objective of the model is to differentiate between actual (live) and false (spoof) faces while maintaining a low computational cost so that it can be implemented on edge devices.

A comparison of commonly used CNN models for face anti-spoofing is shown in Table 3.1. The values are based on trends reported in the literature [8, 9, 13].

Table 3.1: Comparison of CNN Models for Face Anti-Spoofing

Model	**Accuracy (%)**	**Parameters (M)**	**Speed (FPS)**
VGG16	95–99	∼138	Low
ResNet50	96–99	∼25	Medium
CNN (Basic)	90–95	∼5–10	Medium
MobileNet	94–98	∼3–5	High

In feature extraction, input images are resized to a fixed dimension of $H \times W \times C$, where H, W, and C represent the height, width, and number of channels of the image. These images are subsequently processed through multiple convolutional layers to extract meaningful hierarchical features such as edges, textures, and facial structures. These learned characteristics are essential for differentiating between genuine (bonafide) faces and spoof attacks such as printed images or replayed videos [31].

The convolution operation used in feature extraction is defined as:

$$Y(i,j) = \sum_{m} \sum_{n} X(i+m, j+n) \cdot W(m,n) \tag{3.2}$$

where:

- $X(i{+}m, j{+}n)$ defines input image pixel value at spatial location $(i{+}m, j{+}n)$.
- $W(m,n)$ shows convolution kernel (filter) weight at position (m,n).
- $Y(i,j)$ shows output feature map at location (i,j).
- m and n define spatial dimensions of the kernel.
- The summation $\sum_m \sum_n$ represents the sliding window operation over local regions.

These features enable the network to learn local spatial patterns by applying learnable filters over the input image, generating feature maps that highlight

important visual structures. Therefore, to improve computational efficiency, the model adopts a lightweight convolution strategy inspired by depthwise separable convolution, which reduces computational complexity while maintaining feature representation capability. This makes the model suitable for real-time face anti-spoofing applications.

After feature extraction using convolutional layers, a non-linear activation function is used to add complexity to the learning process. This enables the model to capture more detailed and discriminative patterns in facial data, which is essential for distinguishing between real and spoofed faces [10].

A commonly used activation function is the Rectified Linear Unit (ReLU), defined as:

$$f(x) = \max(0, x) \tag{3.3}$$

where x represents the input feature and $f(x)$ represents the activated output. This function retains positive values while suppressing negative values, thereby improving computational efficiency and mitigating the vanishing gradient problem.

After activation, a pooling operation is implemented to reduce the spatial dimensions of the feature maps. This helps decrease computational cost and reduces the risk of overfitting. A commonly used method is max pooling, defined as:

$$Y(i, j) = \max_{(m,n)\in R} X(i + m, j + n) \tag{3.4}$$

where X represents the input feature map, Y represents the output pooled feature map, and R denotes the pooling window region. The maximum value within each region is selected to preserve the most significant features while reducing dimensionality.

The extracted features are passed to a fully connected layer, which performs the final classification. The model generates output probabilities for two classes: real and spoof. Based on these probabilities, the system decides whether the input face is real or fake.

A suitable loss function is used to measure the difference between the

predicted and actual results. The model parameters are updated using an efficient optimization algorithm that helps achieve faster convergence and stable learning. Here, a hybrid learning model is used by combining both image and video data. This model learns spatial features from images and temporal information from videos, improving its overall performance and robustness.

Overall, the model is designed to be simple, fast, and effective, making it suitable for real-time face anti-spoofing applications while maintaining high accuracy.

Table 3.2: Model Hyperparameters

Parameter	**Value**
Model	MobileNet-based CNN
Input Size	$160 \times 160 \times 3$
Batch Size	32
Epochs	5
Learning Rate	0.0001
Optimizer	Adam
Loss Function	Cross-Entropy

The main hyperparameters used to train the proposed model are shown in Table 3.2. These values have been carefully chosen to achieve a balance between computational efficiency and performance. The MobileNet-style CNN, which is known for its lightweight design, serves as the model's foundation. The $160\times160\times3$ input size is sufficient for capturing facial features while maintaining processing speed.

The model uses a batch size of 32 to ensure stable training without requiring excessive memory. The model is trained for 5 epochs, which is sufficient for convergence based on the adopted architecture and preprocessing steps. A learning rate of 0.0001 is selected to allow smooth and gradual updates of the model parameters while reducing the possibility of unstable learning.

The Adam optimizer is used to provide efficient and reliable convergence during training. Finally, cross-entropy loss is selected as the loss function since it is well-suited for classification tasks such as distinguishing between real and spoof faces.

Therefore, the selected hyperparameters help the model achieve a balance between speed, accuracy, and suitability for real-time deployment.

3.4.3 Face Recognition Using Pre-trained SFace

For live faces, the next stage is face recognition. The proposed system uses the **pre-trained SFace face-recognition model** provided through OpenCV. SFace generates a numerical feature representation, or *face embedding*, from the detected and aligned face. OpenCV's implementation supports feature extraction and similarity matching using cosine similarity or distance-based methods [20].

Unlike the anti-spoofing model, SFace is **not retrained using the project's face dataset**. Instead, the system uses SFace to generate embeddings for enrolled persons and subsequently compares the embedding of a newly detected face with the stored embeddings.

For a detected live face F_t, SFace generates an embedding as follows:

$$E_t = \text{SFace}(F_t) \tag{3.5}$$

Similarly, an enrolled person's stored face image is converted into an embedding:

$$E_i = \text{SFace}(F_i) \tag{3.6}$$

The similarity between the two embeddings is calculated using cosine similarity:

$$S(E_t, E_i) = \frac{E_t \cdot E_i}{\|E_t\| \, \|E_i\|} \tag{3.7}$$

where E_t represents the embedding of the currently detected live face, E_i represents the stored embedding of the i-th enrolled person, and $S(E_t, E_i)$ represents the cosine similarity between the two embeddings.

In the current implementation, the generated embeddings are normalized. Therefore, the dot product between the normalized embeddings is equivalent to their cosine similarity. The application uses a recognition threshold of 0.45 for the current implementation.

Accordingly, the identity decision is defined as:

$$\text{Identity} = \begin{cases} \text{Known}, & S(E_t, E_i) \geq 0.45, \\ \text{Unknown}, & S(E_t, E_i) < 0.45. \end{cases} \tag{3.8}$$

It is important to note that 0.45 is the recognition threshold used in the current implementation and is not a universal SFace threshold. The OpenCV SFace implementation also supports cosine-similarity-based matching, with its example implementation using a different cosine threshold. Therefore, the threshold in the proposed system is treated as an application-specific parameter that can be calibrated using validation data [20].

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